RPi Keeb
(Part 1)

Building your own custom keypad is easier than you'd have imagined.

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With any software or service, as you get more comfortable and understand what your workflow pipeline is like, you will notice that you don't need to use more of the functions. And you'll also notice that some functions will be used a lot more often than others. Take for example, editing videos using Adobe Premiere Pro or DaVinci Resolve or any other popular software. If you're editing the

And if these keys are not exactly close to each other, then you'll find it a little cumbersome to work. You can obviously remap the keys but then any other software that's being used at the exact same time might have issues. Often, folks prefer to get a mini keypad that has just a few keys which can be separately remapped to the most popularly used keys. Or ... you can make one yourself using a Raspberry

Raspberry Pi Pico

videos primarily, then the select, cut and delete tools will be used the most often. On the other hand, if you're colour grading, then you'll be spending a lot of time within the Adobe Lumetri tool panel. The former use case needs 3-4 buttons and a little bit of fast scrobbling for which a rotary encoder such as a mouse scroll wheel will be very useful. For the latter use case, you'll need a lot more rotary encoders or even a trackball mouse which is essentially two encoders placed at a right angle. What we're getting at is that you'll whittle down your interaction with the software down to a few keys.

Pi board. Since the Raspberry Pi boards have a very standardised structure, you can follow these steps for any of the Raspberry Pi boards that are currently available. You'll also have to machine a custom keyboard chassis and a PCB for the keypad, but that's for another day.

Here's everything you'll need to get a keyboard working properly:

1. Raspberry Pi (Any of the following models)

a. Pi 1 Model B (2012)
b. Pi 1 Model A (2013)
c. Pi 1 Model B+ (2014)
d. Pi 1 Model A+ (2014)
e. Pi 2 Model B (2015)
f. Pi Zero (2015)
g. Pi 3 Model B (2016)
h. Pi Zero W (2017)
i. Pi 3 Model B+ (2018)
j. Pi 3 Model A+ (2019)
k. Pi 4 Model A (2019)
l. Pi 4 Model B (2020)

2. Breadboard or PCB Board
3. Male-to-Female jumper wires
4. Tactile or Mechanical Switches
5. XY-Joystick module
6. WS2812B LED Module
7. Rotary Encoders
8. 4148 Diodes
9. Software: Raspbian or any other Linux distro

Instructions:

Step 1: Button / Switch Placement

On the breadboard or the PCB, build a matrix using the tactile switches or mechanical switches. We use a matrix because it's easier to address a large number of switches using this method. If you've ever used Excel or any other spreadsheet software then you'll know that each cell is addressed by a combination of the row number and column number. We follow the same logic for switches. Here's what the wiring diagram will look like.

Now, in a separate section of the breadboard or the PCB, place the rotary encoders and the XY-Joystick.

Step 2: Wiring

We now have a lot of rows and columns for the buttons, an XY-joystick module and a couple of rotary encoders. The Raspberry Pi has 26 GPIO (General Purpose Input Output) pins which will be used to address all these devices. Let's say we're using a 4x5 matrix for the buttons. The diagram given above was for a 4x4 matrix, you can add another row at the top or bottom. So we will be connecting the devices in the following order:

- Columns - GP0, 1, 2, 3
- Rows - GP4, 5, 6, 7, 8
- Rotary Encoder 1 - GP9, 10, 11
- Rotary Encoder 2 - GP12, 13, 14
- Joystick Module - GP18, 19, 20